Assignment 4:

Write a C program that takes an integer input representing a month (1 to 12) and a year. Use a switch statement to display the number of days in that month, considering leap years.

Source Code:

```
#include<stdio.h>
int main() {
   int month, year, days;
   printf("Enter the month (1 to 12) and the year : ");
   scanf("%d%d",&month,&year);
   if ((month<1)||(month>12)||(year<0001)) {
      printf("Please enter a positive integer number which is less than 13. ");
      return 1;
   }
   switch (month) {
      case 2 :
         if ((year%4==0 && year%100!=0) || (year%400==0))
            days = 29;
         else
            days = 28;
         break;
      default:
         if ((month==4)||(month==6)||(month==9)||(month==11))
            days = 30;
         else
            days = 31;
   }
   printf("The number of days in that month is %d",days);
   return 0;
}
```

Output:

```
Enter the month (1 to 12) and the year :2

2020

The number of days in that month is 29
Process finished with exit code 0
```

```
Enter the month (1 to 12) and the year :8

1947

The number of days in that month is 31
Process finished with exit code 0
```